

Sprint 6 Review Report

Project: Music Course Platform

Sprint: Sprint 6 (Week 12–13)

Date: 13.4.2026

Scrum Master: Su Wai Phyoe

Team Members: Luo Ying, Lu Liu, Su Wai Phyoe, Yicheng Chen

Sprint Goal

Extend localization support into the database layer while improving overall code quality through static code analysis and refactoring. Prepare the system for acceptance testing and update system architecture documentation.

Key Objectives

- Implement database-level localization with UTF-8 encoding support
- Design and update database schema for multilingual data handling
- Conduct statistical code review using SonarQube / SonarScanner
- Identify code complexity, duplication, and maintainability issues
- Refactor and clean up existing codebase based on analysis results
- Design acceptance test plan (functional, usability, performance testing)
- Update system architecture (ER diagram + UML class diagram)
- Generate missing i18n localization entries from resource files

Completed Tasks

- Database Localization Implementation – Completed by Yicheng Chen

Implemented database localization strategy with UTF-8 support to ensure multilingual data storage and retrieval across the system

- Database Schema Update – Completed by Yicheng Chen

Updated database structure to support multilingual fields and localization requirements

- Statistical Code Review – Completed by Su Wai Phyoe

Performed static analysis using SonarQube/SonarScanner to evaluate cyclomatic complexity, duplication, and code quality issues

- Code Clean-Up and Refactoring – Completed by Yicheng Chen

Refactored complex methods, removed duplicate logic, and improved readability and maintainability of the codebase

- Acceptance Test Plan Design – Completed by Su Wai Phyoe

Designed functional, usability, and performance test cases without execution, mapped to system requirements

- Documentation Update – Completed by Lu Liu
Updated README with localization method explanation and maintained project documentation consistency.
- Architecture Design Update – Completed by Luo Ying
Updated ER diagram (including INSTRUMENT table) and revised UML class diagram according to system changes

Team Contribution Summary

Team Member	Assigned Tasks	Time Spent (hrs)	In class tasks
Luo Ying	ER diagram update, UML diagram update, documentation	12 hrs	Submitted
Lu Liu	README update, localization documentation	13 hrs	Submitted
Su Wai Phyoe	Code review report, acceptance test plan, sprint	13 hrs	Submitted
Yicheng Chen	Database localization, refactoring, implementation	14 hrs	Submitted

Challenges and Solutions

- Database Localization Complexity
Solved by implementing UTF-8 encoding and structured multilingual database design
- High Cyclomatic Complexity in Code
Resolved by splitting large methods into smaller reusable components
- Code Duplication Issues
Removed redundant logic and improved service-layer structure
- Test Design Coverage
Ensured full coverage of functional, usability, and performance test cases without execution

Conclusion

Sprint 6 successfully extended localization into the database layer and significantly improved code quality through static analysis and refactoring. The system is now more maintainable, structured, and ready for final acceptance testing.